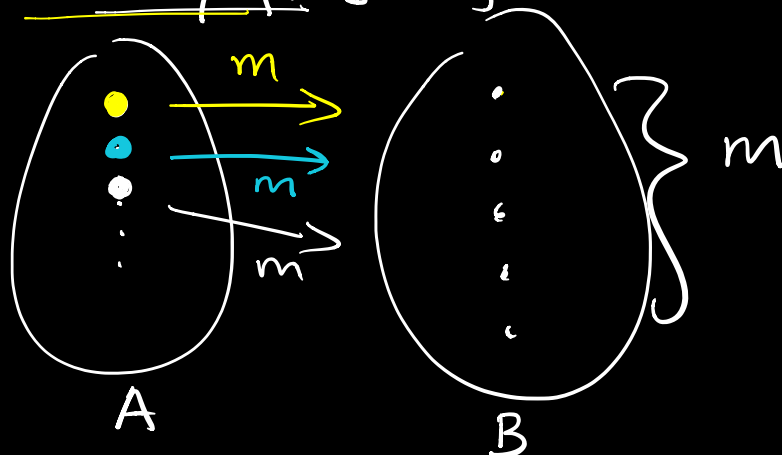


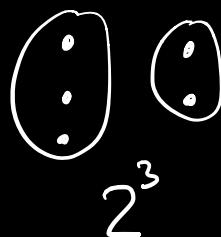
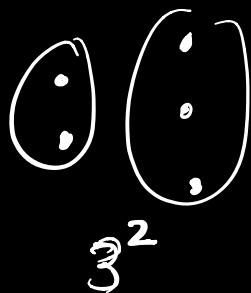
Lesson 25 Counting Functions

$$f: A \rightarrow B, \quad |A| = n, \quad |B| = m$$

① How many fn's $f: A \rightarrow B$?



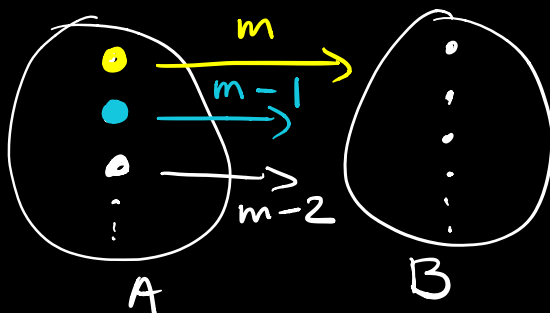
$$\# \text{ of functions} = \underbrace{m \cdot m \cdot \dots \cdot m}_n = m^n$$



② How many 1-to-1 fns?

$$f: A \rightarrow B, \quad |A| = n, \quad |B| = m$$

$n \leq m$:



$$\# \text{ of 1-to-1 fns} = \underbrace{m(m-1)(m-2) \dots (m-n+1)}_n$$

$$\begin{pmatrix} \circ \\ \circ \\ \circ \end{pmatrix} \begin{pmatrix} \circ \\ \circ \\ \circ \\ \circ \end{pmatrix} = \underline{P(m, n)}$$

$$5(4)(3) = P(5, 3)$$

$$= 60$$

$n > m$: no 1-to-1 fn's (PHP)

③ How many onto fn's?

$$f: A \rightarrow B, \quad |A| = n, \quad |B| = m$$

$n < m$: no onto fn's

$$\boxed{n \geq m} :$$

$$B = \{b_1, b_2, \dots, b_m\}$$

$$B_i = \{ \text{fn's which do not hit } b_i \} \quad i=1, 2, \dots, m$$

$$B_1 \cup B_2 \cup \dots \cup B_m = \{ \text{fn's which are not onto} \}$$

$$|B_1 \cup B_2 \cup \dots \cup B_m| = \sum |B_i| - \sum |B_i \cap B_j|$$

$$\text{(inc-exc)} \quad + \sum |B_i \cap B_j \cap B_k| - \dots$$

$$+ (-1)^{m+1} |B_1 \cap B_2 \cap \dots \cap B_m|$$

$$|B_i| = (m-1)^n$$

$$\sum |B_i| = m(m-1)^n$$

$$|B_i \cap B_j| = (m-2)^n$$

$$\sum |B_i \cap B_j| = \binom{m}{2} (m-2)^n$$

↑

of 2-set intersections
from m sets

$$|B_1 \cup B_2 \cup \dots \cup B_m| = m(m-1)^n - \binom{m}{2}(m-2)^n + \binom{m}{3}(m-3)^n - \dots + \dots$$

of onto fn's = total - non-onto

$$= \binom{m}{0} m^n - \binom{m}{1} m(m-1)^n + \binom{m}{2}(m-2)^n - \binom{m}{3}(m-3)^n + \dots$$

$$= \sum_{i=0}^{m-1} (-1)^i \binom{m}{i} (m-i)^n$$

Notes : Numerical example

HW 9 #1 PHP

7 random integers

Show : There are 2 s.t.
their sum or difference
is a multiple of 10

0	
1, 9	19
2, 8	
3, 7	43

(remainder
when divided
by 10 :
0, 1, 2, ..., 9)

$$\frac{4, 6}{5}$$

$$\underline{\underline{r}}, \underline{\underline{10-r}}$$

By PHP there must be 2 integers a, b
in the same hole

• a, b have same remainder r

$$a = 10n + r$$

$$b = 10m + r$$

$$(a - b) = \dots = \underline{\underline{10}}(\quad)$$

• a has remainder r,
b has ... 10-r

$$a = 10n + r$$

$$b = 10m + (10 - r)$$

$$(a + b) = \dots = \underline{\underline{10}}(\quad)$$